CAPSTONE TWO | SPRINGBOARD DATA SCIENCE

Where You Grow Up Matters

Predicting adult income for low-income children in the DC region

Brittany Barrion



Mural by Lula Goce, Washington, DC

The question

Which neighborhood characteristics are most closely tied to adult income for children from low-income families, and can they help decide where to focus first?

Who it is for

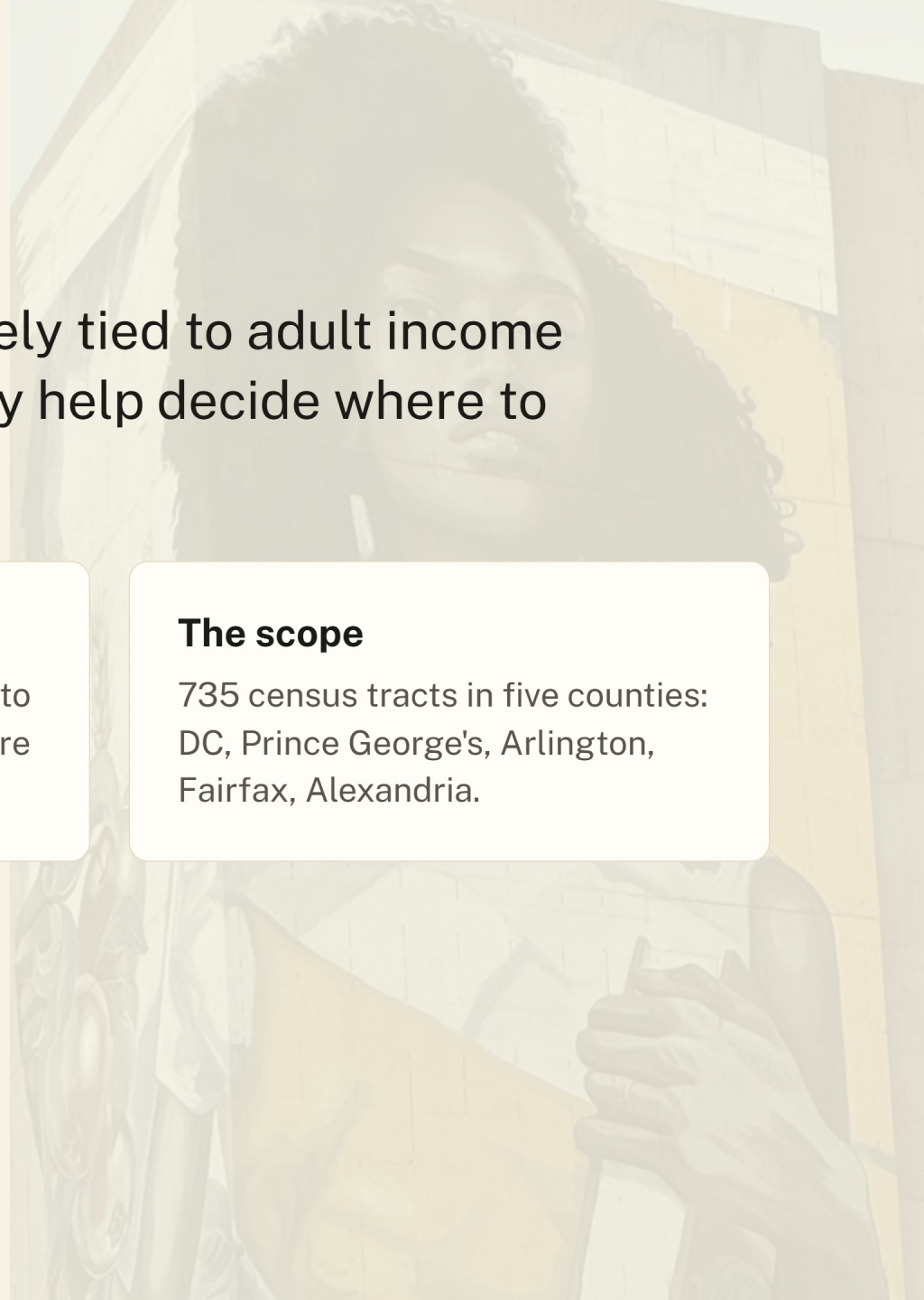
A DC-area foundation and a community development agency deciding where to invest.

The outcome

Adult household income rank (0 to 1) for children whose parents were at the 25th percentile.

The scope

735 census tracts in five counties: DC, Prince George's, Arlington, Fairfax, Alexandria.



Approach

1. Wrangle

Joined two national files, kept five counties, dropped sparse columns and tiny tracts.

2. Explore

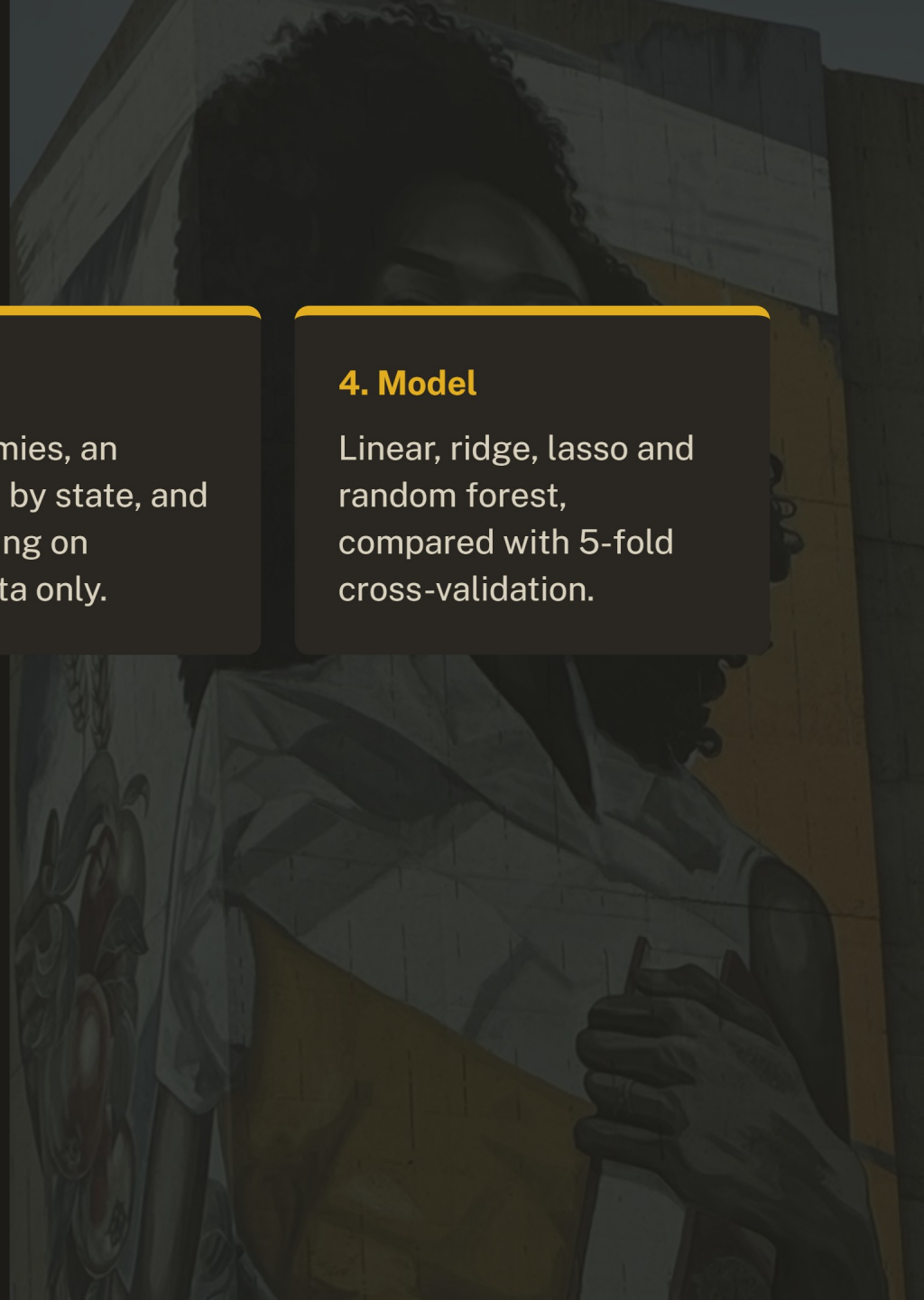
Correlations, state comparisons, and overlap checks between predictors.

3. Prepare

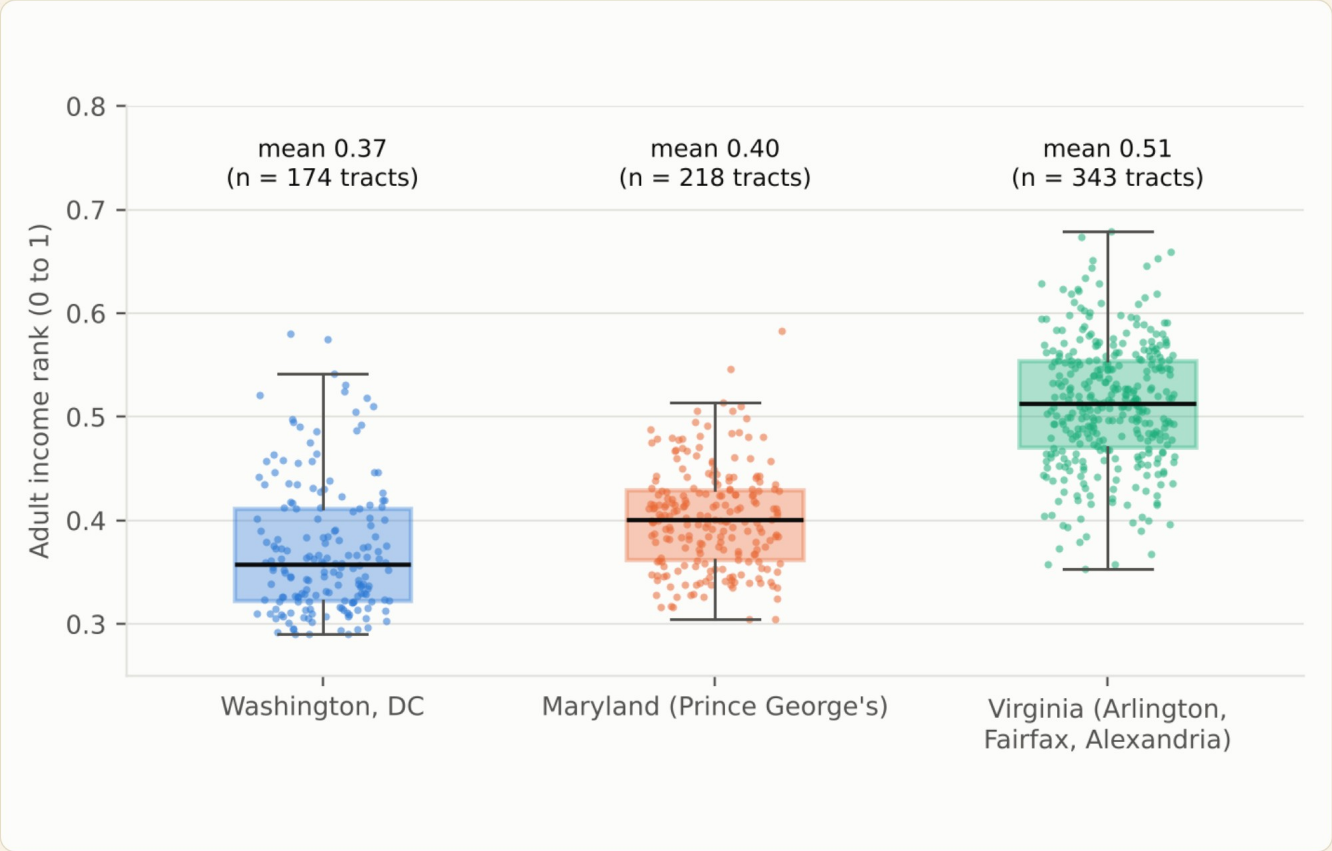
State dummies, an 80/20 split by state, and standardizing on training data only.

4. Model

Linear, ridge, lasso and random forest, compared with 5-fold cross-validation.



Outcomes differ sharply across the region

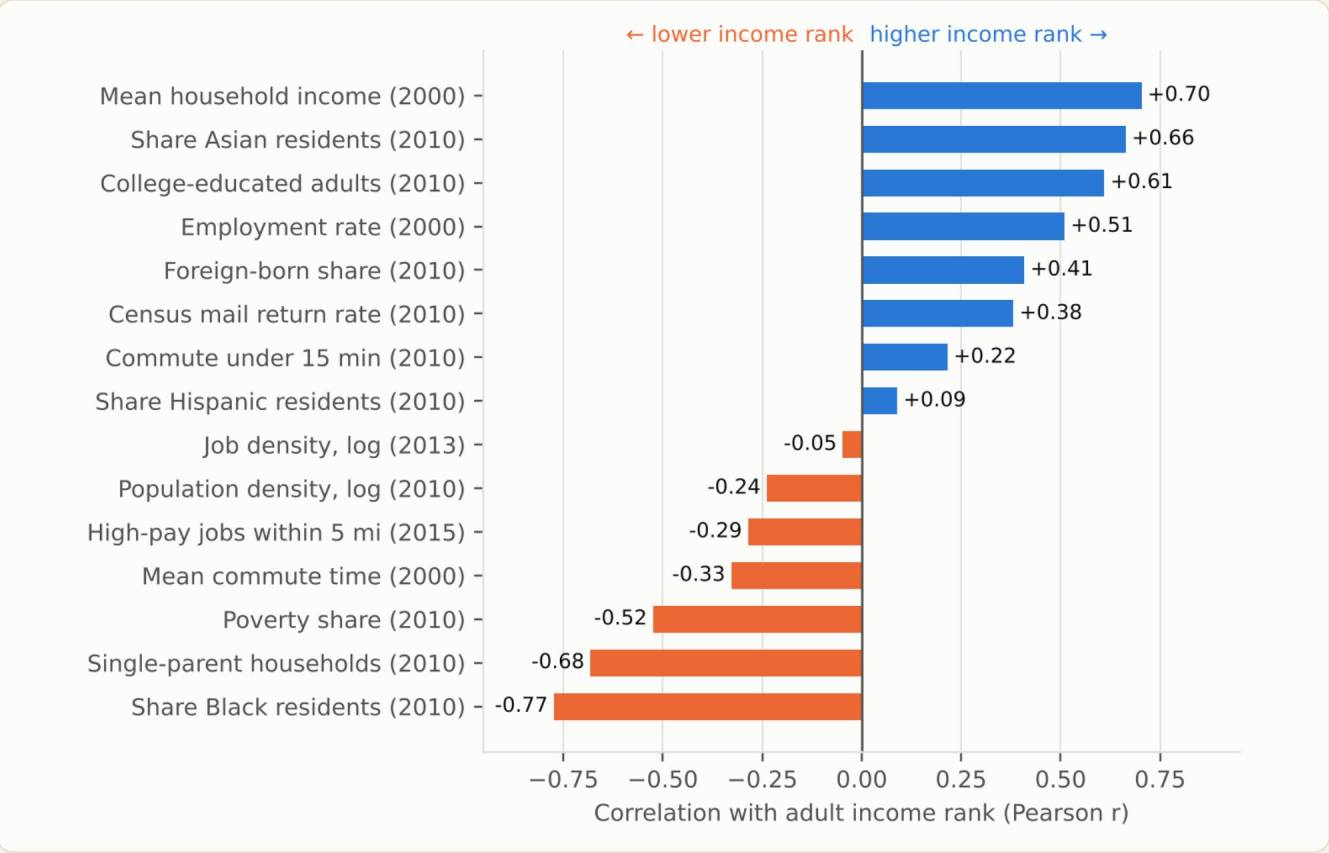


0.51

Average rank in Northern Virginia, versus 0.40 in Prince George's and 0.37 in DC.

Spread within each area is wide, and the three groups still overlap.

Income, education and family structure track outcomes



Higher outcomes: past household income (+0.70), share Asian (+0.66), college-educated adults (+0.61).

Lower outcomes: share Black (-0.77), single-parent households (-0.68), poverty (-0.52).

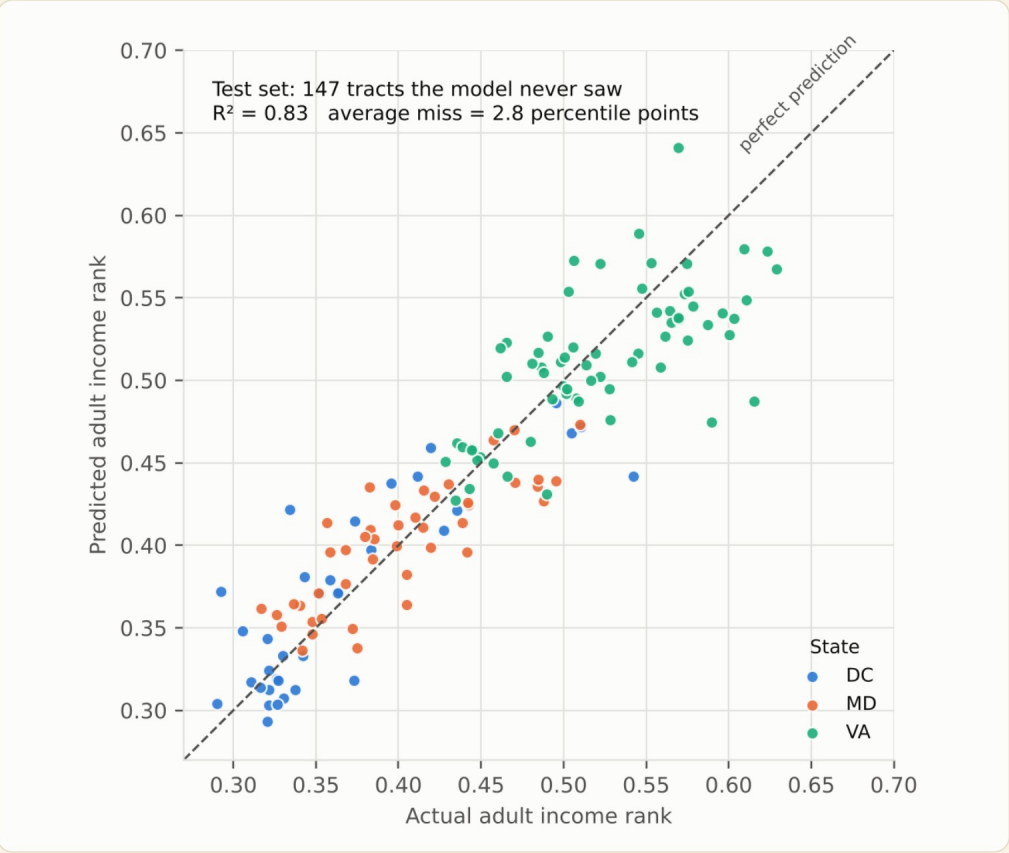
Associations, not causes.

Simple model, nearly the best score

Model	CV R^2	CV MAE	Test R^2	Test MAE
Baseline (average)	-0.029	7.1	-0.001	7.6
Linear regression (final)	0.745	3.1	0.833	2.8
Ridge regression	0.747	3.0	0.826	2.9
Lasso regression	0.749	3.0	0.829	2.9
Random forest (tuned)	0.757	2.9	0.846	2.7

The forest leads by 0.012 R^2, less than the 0.014 standard error, so the one-standard-error rule picks the simplest model. MAE is in percentile points.

Predictions land close to reality



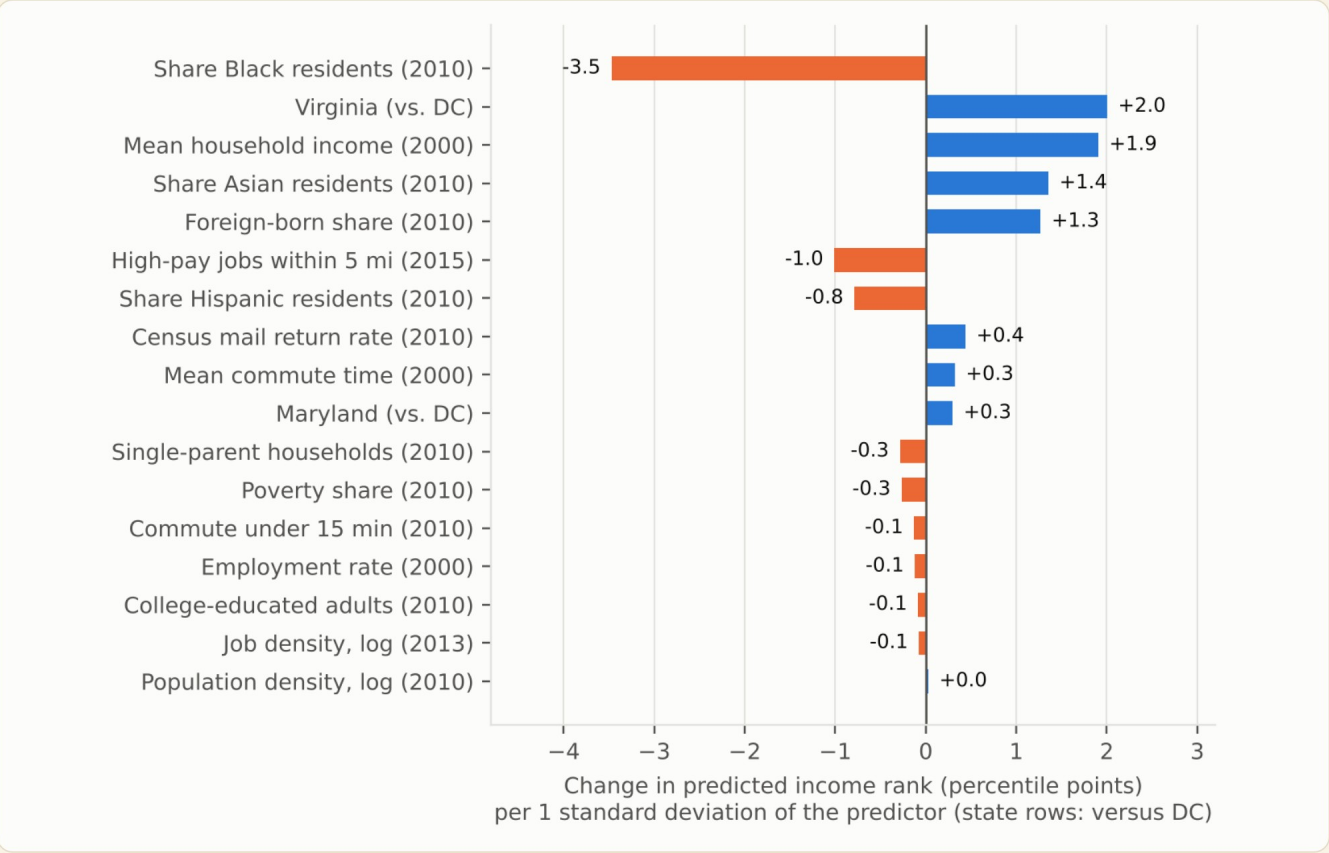
2.8 pts

Average miss on 147 unseen tracts, versus 7.1 for guessing the average.

33 of the 37 tracts in the actual bottom quarter were also in the model's bottom quarter.

Cross-validated R^2 is 0.745, the more cautious figure.

What the final model relies on



Largest weights: share Black, household income (2000), share Asian, foreign-born share.

Overlapping predictors share credit, so read the pattern, not single bars.

State adds little: test R^2 is 0.824 without it, 0.833 with it.

Three recommendations

1. Screen, don't judge

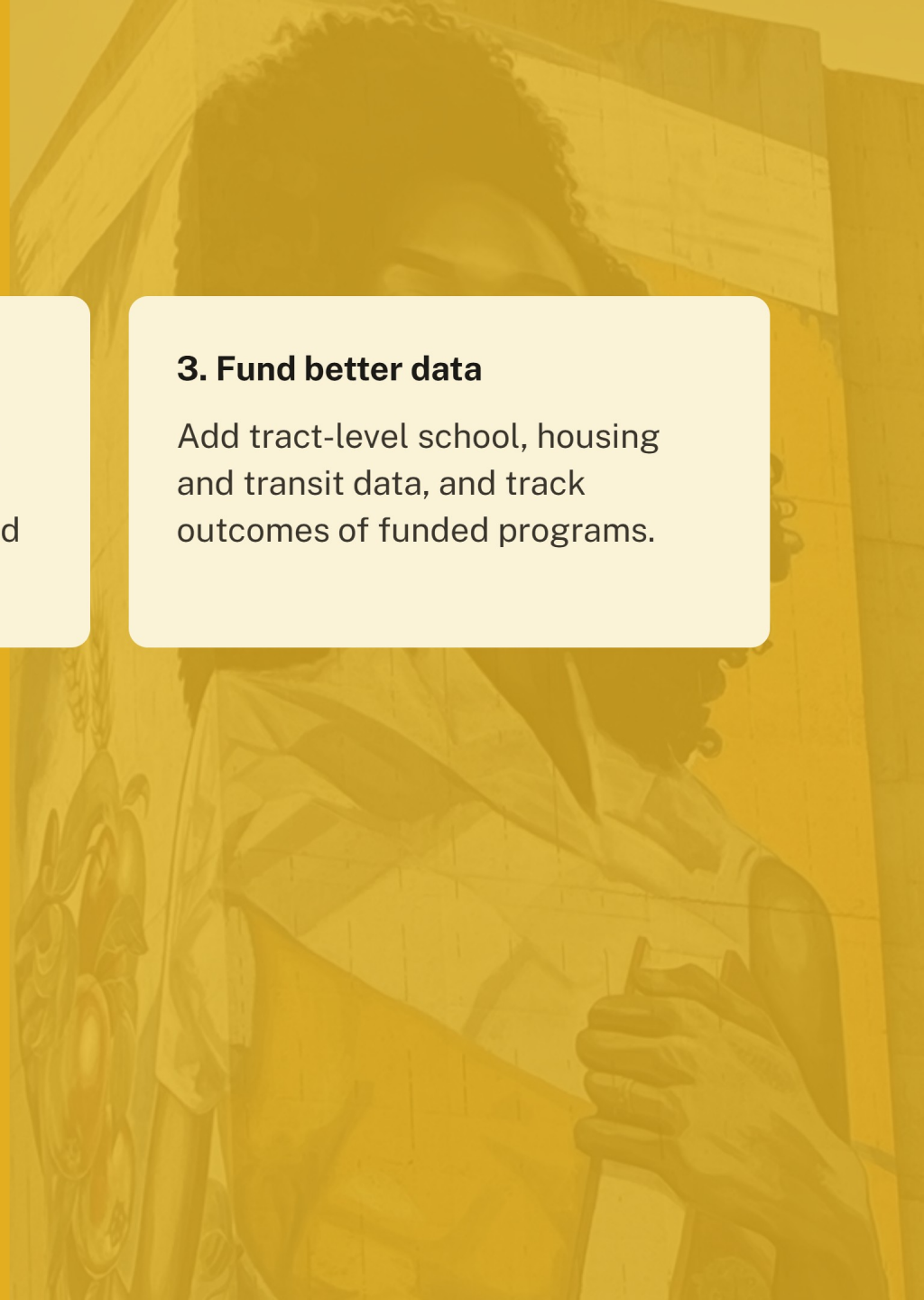
Rank tracts by predicted outcome to decide where to look first. Never use it to decide eligibility for any household.

2. Invest in places

Pilot workforce, adult education, small-business and housing support in low-scoring tracts, and measure what changes.

3. Fund better data

Add tract-level school, housing and transit data, and track outcomes of funded programs.



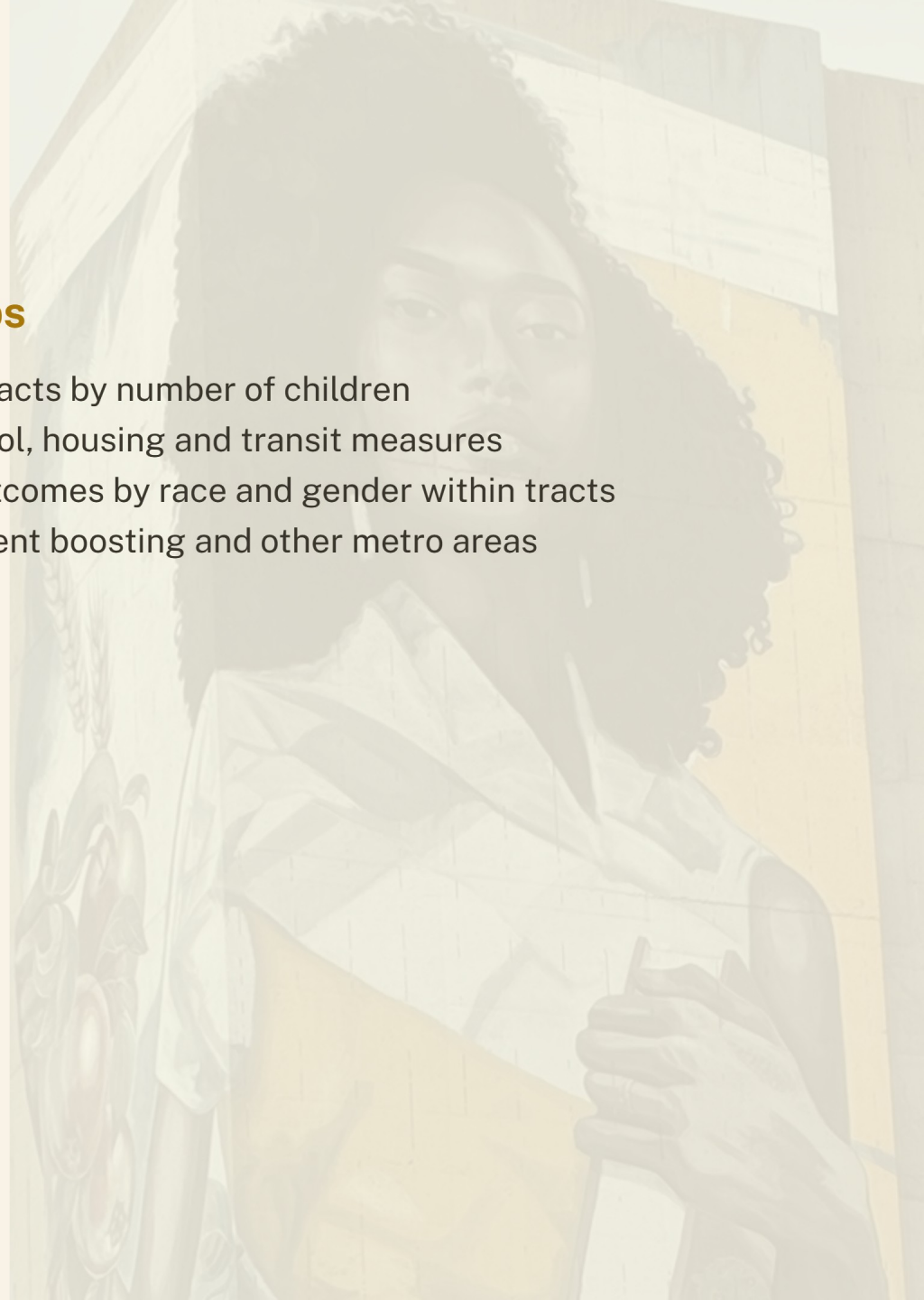
Limits and next steps

Limits

- Associations at the neighborhood level, not causes
- Only 735 tracts and 147 test tracts
- Outcomes (born 1978 to 1983) predate predictors (2000 to 2015)
- Overlapping predictors make single weights unstable

Next steps

- Weight tracts by number of children
- Add school, housing and transit measures
- Study outcomes by race and gender within tracts
- Try gradient boosting and other metro areas



Opportunity is shaped by place.

A simple model can show where to look first. People and programs on the ground show what to do next.

Thank you | Questions



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